Generating fn:

$\{a_n\}$, $A(s) = \sum_0^\infty a_n s^n$

Prob. gen. fn X, $a_n = P(X=n)$

$$P_X(s) = \sum_{n=0}^{\infty} P(X=n) s^n.$$

Ex 1 Coin is tossed n times.

Find the prob of number of heads being even.

Denote $\epsilon_i = \begin{cases} 1 & \text{if } i\text{th coin toss in H} \\ 0 & \text{o.w} \end{cases}$

Define $S_n = \sum_{1}^{n} \epsilon_i, \; n \geq 1$, $S_0 = 0$.

$\boxed{P(\epsilon_i = 1) = p}$

$p_n = P(S_n \text{ is even})$

$$p_0 = 1$$

$$p_1 = P(\epsilon_1 \text{ is even}) = P(\epsilon_1 = 0) = 1-p$$

$$p_2 = P(\epsilon_1+\epsilon_2 \text{ is even}) = P(\epsilon_1+\epsilon_2=0) + P(\epsilon_1+\epsilon_2=2)$$

$$= P(\epsilon_1=0, \epsilon_2=0) + P(\epsilon_1=1, \epsilon_2=1)$$

$$= P(\epsilon_1=0)P(\epsilon_2=0) + P(\epsilon_1=1)P(\epsilon_2=1)$$

$$= p^2 + (1-p)^2$$

Recurrence relation

depends $(\epsilon_1 \cdots \epsilon_n)$

$$S_{n+1} = \underbrace{\epsilon_1 + \cdots + \epsilon_n}_{S_n} + \epsilon_{n+1}$$

$$= S_n + \epsilon_{n+1}$$

- $p_0 = 1$,

- $p_{n+1} = P(S_{n+1} \text{ is even})$

$$= P(S_{n+1} \text{ is even}, \epsilon_1 = 1)$$

$$+ P(S_{n+1} \text{ is even}, \epsilon_1 = 0)$$

$$= P(S_{n+1} - S_1 \text{ is odd}, \epsilon_1 = 1)$$

$$+ P(S_{n+1} - S_1 \text{ is even}, \epsilon_1 = 0)$$

$$= P(S_{n+1} - S_1 \text{ is odd}) P(\epsilon_1 = 1)$$

$$+ P(S_{n+1} - S_1 \text{ is even}) P(\epsilon_1 = 0)$$

$$= P(S_n \text{ is odd}) P(\epsilon_1 = 1)$$

$$+ P(S_n \text{ is even}) P(\epsilon_1 = 0)$$

$$= (1 - p_n) p + p_n (1 - p)$$

$S_n = \underbrace{\epsilon_1 + \cdots + \epsilon_n}$

$S_{n+1} - S_1 = \epsilon_2 + \cdots + \epsilon_{n+1}$

$S_n \overset{d}{=} S_{n+1} - S_1$

$\underline{n=0}$, $p_1 = (1 - \cancel{p_0}) p + p_0 (1-p)$

$= 1 - p$

$$X \overset{d}{=} Y \iff P(X \in A) = P(Y \in A) \quad \forall A$$

$$\epsilon_1, \dots, \epsilon_{n+1} \text{ iid}$$

$$\epsilon_1 + \epsilon_2 \overset{d}{=} \epsilon_3 + \epsilon_4 \qquad f : \mathbb{R}^k \longmapsto \mathbb{R}$$

$$f : \mathbb{R}^2 \longmapsto \mathbb{R}, \; f(u,v) = u+v \qquad f(\epsilon_1, \dots, \epsilon_k) \overset{d}{=} f(\epsilon_{i_1}, \dots, \epsilon_{i_k})$$

$p_0 = 1, \quad p_{n+1} = (1-p_n)p + p_n(1-p) \quad \forall\, n \geq 0$

$$P(s) = \sum_{n=0}^{\infty} p_n s^n$$

$$\sum_{n=0}^{\infty} p_{n+1} s^{n+1} = \sum_{n=0}^{\infty} (1-p_n) p s^{n+1} + \sum_{n=0}^{\infty} p_n (1-p) s^{n+1}$$

$$\Rightarrow P(s) - 1 = ps \sum_{n=0}^{\infty} s^n - ps \sum_{n=0}^{\infty} p_n s^n + (1-p)s \sum_{0}^{\infty} p_n s^n$$

$$= \frac{ps}{1-s} - sp\,P(s) + (1-p)s\,P(s)$$

$$P(s) - 1 = \frac{ps}{1-s} - spP(s) + s(1-p)P(s)$$

$$\Rightarrow P(s) = \frac{ps - s + 1}{(1-s)(1-s+2sp)}$$

$$= \begin{cases} \dfrac{1-\frac{s}{2}}{1-s} & p = 1/2 \\[2ex] \dfrac{ps-s+1}{(1-s)(1-s+2ps)} & p \neq 1/2 \end{cases}$$

$$\frac{\left(1-\frac{s}{2}\right)}{1-s} = \sum_{n=0}^{\infty} s^n - \frac{1}{2}\sum_{n=0}^{\infty} s^{n+1}$$

$$= \sum_{0}^{\infty} s^n - \frac{1}{2}\sum_{n=1}^{\infty} s^n = 1 + \sum_{n=1}^{\infty} \frac{1}{2} s^n \Rightarrow p_0 = 1, \ p_n = 1/2 \ \forall n \geq 1$$

$p \neq 1/2$

$$P(s) = \frac{1-(1-p)s}{(1-s)(1+s(2p-1))}$$

$$\frac{1}{1-\alpha s} = \sum_{n=0}^{\infty} \alpha^n s^n$$

$$= \frac{A}{1-s} + \frac{B}{1+s(2p-1)}$$

$$= \sum_{n=0}^{\infty} A s^n + B \sum_{n=0}^{\infty} s^n (1-2p)^n$$

$$1-(1-p)s = A(1+s(2p-1)) + B(1-s)$$

$1 = A + B$

$1-p = A(2p-1) - B$

$$\frac{B}{1-s(1-2p)}$$

$P_n = A + B(1-2p)^n \rightarrow \frac{1}{2}$, as $n \to \infty$

$A = 1/2$, $B = 1/2$

Ex 2 :- Toss a coin until two consecutive Head appears. Let p_n be the prob that expt. stops after n tosses.

$n=7$

T H T H T H H

$p_0 = 0$

$p_1 = 0$,

$p_2 = p^2$

E_n = Expt. stops at n; fix $n \geq 3$

$$P(E_n) = P(E_n, \epsilon_1 = 1) + P(E_n, \epsilon_1 = 0)$$

$$= P(\epsilon_1 = 0)\, P(E_{n-1})$$

$$+ \cancel{P(E_n, \epsilon_1 = 1, \epsilon_2 = 1)}$$

$$+ P(E_n, \epsilon_1 = 1, \epsilon_2 = 0)$$

$$= (1-p)\, p_{n-1} + p(1-p)\, P(E_{n-2})$$

$$\boxed{p_n = (1-p)\, p_{n-1} + p(1-p)\, p_{n-2}.} \quad \forall\, n \geq 3$$

$n-1$ n

H ? - - - - . HH

HT . . . - - - - HH

$$\sum_{k=0}^{\infty} p_k = \lim_{n\to\infty} \sum_{k=0}^{n} p_k = 1$$

$P(T = +\infty) = 0$

as $n \to \infty$ $\quad P(T \leq n) \to 1 \Rightarrow P(T > n) \to 0.$

$$P(s) = \sum_{n=2}^{\infty} p_n s^n$$

$$= p_2 s^2 + \sum_{n=3}^{\infty} p_n s^n$$

$$= p_2 s^2 + \sum_{n=3}^{\infty} (1-p) p_{n-1} s^n + p(1-p) \sum_{n=3}^{\infty} p_{n-2} s^n$$

$$= p_2 s^2 + (1-p)s \sum_{n=2}^{\infty} p_n s^n + (1-p) p s^2 \sum_{n=1}^{\infty} p_n s^n$$

$$= p_2 s^2 + (1-p) s P(s) + (1-p) p s^2 P(s)$$

$a_n \geq 0$,

$A(s) = \sum a_n s^n$.

$A(1) = \lim_{s \uparrow 1} A(s) = \underline{\underline{\sum_{n \geq 0}^{\infty} a_n}}$

$$\boxed{\begin{aligned} P(1) &= p_2 + (1-p) P(1) + p(1-p) P(1) \\ \Rightarrow P(1) &= 1 \end{aligned}}$$

$$\sum_{n \geq 0}^{\infty} p_n = P(1) = \underline{1}.$$

Let T_{HH} be the no. of tosses to get 2 consequtive H.

Then, $P(T_{HH} = n) = p_n$.

and T is a proper r.v. $\boxed{P(T=\infty)=0}$

Calculate $E(T_{HH}) = P(\, 1 \,)$.

Let T_{HT} be the no. of tosses req'd to get T followed by H for the first time.

$$E(T_{HH}) = 6 \qquad E(T_{HT}) = 4 \qquad \underbrace{HHHHHHH\cdots HHT}_{n}$$

$$q_n = P(T_{HT} = n), \quad n \geq 3.$$

$$= P(T_{HT} = n, \epsilon_1 = 0) + P(T_{HT} = n, \epsilon_1 = 1)$$

$$= q\, q_{n-1} + p^{n-1} q, \quad q = 1 - p.$$